Problem 27.1

A 20-year increasing annuity due pays 100 at the start of year 1, 105 at the start of year 2, 110 at the start of year 3, and so on. In other words, each payment is \$5 greater than the prior payment. Calculate the present value of this annuity at an annual effective rate of 5%.

Solution

Index the payments by $t = 0, 1, \ldots, 19$, so the payment at time t is $100 + 5t$. Let $v = 1/1.05$ and $d = 1 - v = 0.05/1.05$. The present value is

$$\begin{aligned} PV &= \sum_{t=0}^{19} (100 + 5t) v^t \\ &= 95 \sum_{t=0}^{19} v^t + 5 \sum_{t=0}^{19} (t+1) v^t \\ &= 95 \left(\frac{1 - v^{20}}{d} \right) + 5 \left(\frac{\frac{1-v^{20}}{d} - 20v^{20}}{d} \right) \\ &\approx 1{,}825.596258. \end{aligned}$$

Therefore, the present value of the annuity is

$$\boxed{1{,}825.60}.$$

Using the BA II Plus Professional

For a faster TVM calculation, write each payment as $95 + 5(t+1)$, where $t = 0, 1, \ldots, 19$. Let $v = 1/1.05$ and $d = 1 - v = 0.05/1.05$. Using $\sum_{t=0}^{19} (t+1) v^t = (\ddot{a}_{20} - 20v^{20})/d$,

$$\begin{aligned} PV &= 95 \ddot{a}_{20} + 5 \left(\frac{\ddot{a}_{20} - 20v^{20}}{d} \right) \\ &= \left(95 + \frac{5}{d} \right) \ddot{a}_{20} - \frac{100}{d} v^{20} \\ &= 200 \ddot{a}_{20} - 2{,}100 v^{20}. \end{aligned}$$

This is equivalent to TVM entries $N = 20$, $I/Y = 5$, $PMT = 200$, and $FV = -2{,}100$, with the calculator in BGN mode. The sequence assumes $P/Y = 1$, since payments are annual.

If `BGN` is not displayed, press `2nd BGN`, `2nd SET`, then `2nd QUIT`. Then enter:

1. `2nd CLR TVM` (clear the TVM registers);
2. `20 N`;

3. `5 I/Y`;
4. `200 PMT`;
5. `2100 +/- FV`;
6. `CPT PV`.

The result is approximately 1,825.60. The 1,826.06 shown in the supplied image does not match these inputs; the correct value is 1,825.5963 before rounding.